

Section A

● Choose correct answer from the given options. [Each carries 1 Mark] [23]

1. If $f(x) = \begin{cases} 1+x^2, & 0 \leq x \leq 1 \\ 1-x, & x > 1 \end{cases}$, then
 (A) $\lim_{x \rightarrow 1^+} f(x) \neq 0$ (B) $\lim_{x \rightarrow 1^-} f(x) \neq 2$
 (C) f is discontinuous at $x = 1$ (D) none of these
2. $\frac{d}{dx} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 = \dots\dots\dots$.
 (A) $1 + \frac{1}{x^2}$ (B) $1 - \frac{1}{x^2}$ (C) $1 + \log x$ (D) $2x - \frac{2}{x^3}$
3. $\frac{d}{dx} (\sin(x^2) + \cos(x^2)) = \dots\dots\dots$
 (A) 0 (B) $2x(\cos(x^2) - \sin(x^2))$ (C) $2 \cos 2x$ (D) $\cos(x^2) - \sin(x^2)$
4. $f(x) = x(\sqrt{x} - \sqrt{x+1})$ then f is
 (A) continuous at $x = 0$ but not differentiable. (B) differentiable at $x = 0$.
 (C) not differentiable at $x = 0$. (D) none of these.
5. If $y = 3x^5 + 4x^4 + 2x + 3$, then
 (A) $y_4 = 0$ (B) $y_5 = 0$ (C) $y_6 = 0$ (D) none of these
6. If $y = \sqrt{(1-x)(1+x)}$, then
 (A) $(1-x^2)\frac{dy}{dx} - xy = 0$ (B) $(1-x^2)\frac{dy}{dx} + xy = 0$
 (C) $(1-x^2)\frac{dy}{dx} - 2xy = 0$ (D) $(1-x^2)\frac{dy}{dx} + 2xy = 0$
7. If $y = \frac{\tan x + \cot x}{\tan x - \cot x}$, then $\frac{dy}{dx} = \dots\dots\dots$.
 (A) $2 \tan 2x \sec 2x$ (B) $\tan 2x \sec 2x$ (C) $-\tan 2x \sec 2x$ (D) $-2 \tan 2x \sec 2x$
8. $\frac{d}{dx} \left[\log \sqrt{\sin \sqrt{e^x}} \right] = \dots\dots\dots$.
 (A) $\frac{1}{4} e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$ (B) $e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$ (C) $\frac{1}{4} e^x \cot (e^x)$ (D) $\frac{1}{2} e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$
9. $\left[\frac{d}{dx} \sec^{-1} x \right]_{x=-3} = \dots\dots\dots$.
 (A) $\frac{1}{\sqrt{x^2-1}}$ (B) $\frac{-1}{\sqrt{x^2-1}}$ (C) $\frac{1}{6\sqrt{2}}$ (D) $-\frac{1}{6\sqrt{2}}$
10. $\frac{d}{dx} \sin^{-1} (\cos 2x) = \dots\dots\dots$. $\left(0 < x < \frac{\pi}{2} \right)$
 (A) 2 (B) -2 (C) 1 (D) 5
11. $x^{\cos y} + y^{\cos x} = 5$ then
 (A) $x = 0, y = 0 \Rightarrow y' = 0$ (B) $x = 0, y = 1 \Rightarrow y' = 0$
 (C) $x = y = 1 \Rightarrow y' = -1$ (D) $x = 1, y = 0 \Rightarrow y' = 1$

12. $\frac{d^n}{dx^n}(\log x) = \dots\dots$
 (A) $\frac{(n-1)!}{x^n}$ (B) $\frac{n!}{x^n}$ (C) $\frac{(n-2)!}{x^n}$ (D) $(-1)^{n-1} \frac{(n-1)!}{x^n}$
13. $\frac{d}{dx}(\tan x^\circ) = \dots\dots\dots$
 (A) $\pi \sec^2 x^\circ$ (B) $\frac{180}{\pi} \sec^2 x^\circ$ (C) $\sec^2 x^\circ$ (D) $\frac{\pi}{180} \sec^2 x^\circ$
14. If $e^y(x+1) = 1$, then what is the value of $\frac{d^2y}{dx^2} = \dots\dots\dots$
 (A) $-\left(\frac{dy}{dx}\right)^2$ (B) $\left(\frac{dy}{dx}\right)$ (C) $-\left(\frac{dy}{dx}\right)$ (D) $\left(\frac{dy}{dx}\right)^2$
15. $f(x) = \begin{cases} kx^2 & \text{if } x \leq 2 \\ 3 & \text{if } x > 2 \end{cases}$ If f is continuous at $x = 2$, then the value of k is $\dots\dots$
 (A) $\frac{4}{3}$ (B) $\frac{3}{4}$ (C) 4 (D) 3
16. $\frac{d}{dx}e^{x^3} = \dots\dots\dots$
 (A) e^{x^3} (B) $x^3 \cdot e^{x^3-1}$ (C) $3x^2e^{x^3}$ (D) $e^{x^3}\log_3 e$
17. If $f(x) = \log_a n x^n$ then $f'(1) = \dots\dots\dots$
 (A) $\frac{1}{x \log a}$ (B) 0 (C) $\frac{1}{\log a}$ (D) $\frac{n}{\log n}$
18. $\frac{d}{dx}(x^{2x}) = \dots\dots\dots$
 (A) $x^{2x}(1 + 2\log x)$ (B) $x^{2x}(2 + \log x)$ (C) $2x^{2x}(1 + \log x)$ (D) does not exist
19. $\frac{d}{dx}(\operatorname{cosec}^{-1} e^x) = \dots\dots\dots$
 (A) $\frac{-1}{\sqrt{e^{2x}-1}}$ (B) $\sin^{-1}(e^x)$ (C) $\frac{1}{\sqrt{e^{2x}-1}}$ (D) $\frac{-e^x}{\sqrt{e^{2x}-1}}$
20. If $y = 100e^{2x} + 200e^{-2x}$ and $\frac{d^2y}{dx^2} = ay$, then $a = \dots\dots\dots$
 (A) 0 (B) -4 (C) 2 (D) 4
21. If $x + 1 = e^{-y}$, then $\frac{d^2y}{dx^2} = \dots\dots\dots$
 (A) $\frac{dy}{dx}$ (B) $\left(\frac{dy}{dx}\right)^2$ (C) $\left(\frac{dy}{dx}\right)^3$ (D) $-\frac{dy}{dx}$
22. If $x = \sqrt{10^{\sin^{-1}t}}$, $y = \sqrt{10^{\cos^{-1}t}}$, then $\frac{dy}{dx} = \dots\dots\dots$
 (A) $-\frac{y}{x}$ (B) $\frac{y}{x}$ (C) 0 (D) $-\frac{x}{y}$
23. $\left\{ \frac{d}{dx}(x^x + x^{x+1} + x^{x+2}) \right\}_{x=e} = \dots\dots\dots$
 (A) $e^e(2e^2 + 4e + 3)$ (B) $e^e(1 + e^2 + 2e)$ (C) $e^e(3e^2 + 2e + 2)$ (D) $e^e(1 + 4e + 2e^2)$

Section B

● Write the answer of the following questions. [Each carries 2 Marks]

[18]

1. Find the values of k so that the function f is continuous at the indicated point in :

$$f(x) = \begin{cases} kx + 1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases} \text{ at } x = \pi$$

2. Examine that $\sin|x|$ is a continuous function.
3. Find $\frac{dy}{dx}$ in the following : $ax + by^2 = \cos y$
4. Find $\frac{dy}{dx}$ in the following : $\sin^2 x + \cos^2 y = 1$
5. Differentiate the following w.r.t. x : $\frac{e^x}{\sin x}$
6. Differentiate the following w.r.t. x : e^{x^3}
7. Differentiate the following w.r.t. x : $e^{\sin^{-1} x}$
8. Differentiate the following w.r.t. x : $e^x + e^{x^2} + \dots + e^{x^5}$
9. Differentiate the function with respect to x : $\cos(\sqrt{x})$

Section C

- Write the answer of the following questions. [Each carries 3 Marks] [9]
10. For what value of λ is the function defined by $f(x) = \begin{cases} \lambda(x^2 - 2x), & \text{if } x \leq 0 \\ 4x + 1, & \text{if } x > 0 \end{cases}$ continuous at $x = 0$? What about continuity at $x = 1$?
 11. Find the values of a and b such that the function defined by $f(x) = \begin{cases} 5 & , \text{ If } x \leq 2 \\ ax + b, & \text{ If } 2 < x < 10 \\ 21 & , \text{ If } x \geq 10 \end{cases}$ is a continuous function.
 12. Find all the points of discontinuity of f defined by $f(x) = |x| - |x + 1|$.

Section A

- Choose correct answer from the given options. [Each carries 1 Mark]

[23]

1. If $f(x) = \begin{cases} 1+x^2, & 0 \leq x \leq 1 \\ 1-x, & x > 1 \end{cases}$, then

(A) $\lim_{x \rightarrow 1^+} f(x) \neq 0$

(B) $\lim_{x \rightarrow 1^-} f(x) \neq 2$

(C) f is discontinuous at $x = 1$

(D) none of these

Ans. (C) f is discontinuous at $x = 1$

➔ Here, $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 0^+} 1 - x$ ($\because f(x) = x - 1, x > 1$)

$= \lim_{h \rightarrow 0} 1 - (1 + h)$

$= 1 - (1 + 0) = 0$

and $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 1 + x^2$

$= \lim_{h \rightarrow 0} 1 + (1 - h)^2$

$= 1 + (1 - 0)^2$

$= 1 + (1^2)$

$= 2$

$\therefore \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x)$

$\therefore \lim_{x \rightarrow 1} f(x)$ does not exist.

$\therefore f(x)$ is discontinuous at $x = 1$.

2. $\frac{d}{dx} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 = \dots\dots\dots$

(A) $1 + \frac{1}{x^2}$

(B) $1 - \frac{1}{x^2}$

(C) $1 + \log x$

(D) $2x - \frac{2}{x^3}$

Ans. (B) $1 - \frac{1}{x^2}$

➔ $\frac{d}{dx} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 = \frac{d}{dx} \left(x + 2 + \frac{1}{x} \right)$

$= \frac{d}{dx} x + \frac{d}{dx} 2 + \frac{d}{dx} \left(\frac{1}{x} \right)$

$= 1 + 0 - \frac{1}{x^2}$

$= 1 - \frac{1}{x^2}$

3. $\frac{d}{dx} (\sin(x^2) + \cos(x^2)) = \dots\dots\dots$

(A) 0

(B) $2x(\cos(x^2) - \sin(x^2))$

(C) $2 \cos 2x$

(D) $\cos(x^2) - \sin(x^2)$

Ans. (B) $2x(\cos(x^2) - \sin(x^2))$

$$\begin{aligned}\Rightarrow \frac{d}{dx} (\sin(x^2) + \cos(x^2)) &= \cos(x^2) \cdot \frac{d}{dx}(x^2) - \sin(x^2) \frac{d}{dx}(x^2) \\ &= 2x \cdot \cos(x^2) - 2x \sin(x^2) \\ &= 2x [\cos(x^2) - \sin(x^2)]\end{aligned}$$

4. $f(x) = x(\sqrt{x} - \sqrt{x+1})$ then f is

- (A) continuous at $x = 0$ but not differentiable. (B) differentiable at $x = 0$.
(C) not differentiable at $x = 0$. (D) none of these.

Ans. (B) differentiable at $x = 0$.

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x(\sqrt{x} - \sqrt{x+1}) = 0$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x(\sqrt{x} - \sqrt{x+1}) = 0$$

$\therefore f(x)$ is continuous at $x = 0$

$$\begin{aligned}Lf'(0) &= \lim_{h \rightarrow 0} \frac{(0-h)(\sqrt{0-h} - \sqrt{0-h+1}) - 0}{-h} \\ &= \lim_{h \rightarrow 0} \sqrt{-h} - \sqrt{-h+1} \\ &= -\sqrt{1} = -1\end{aligned}$$

$$\begin{aligned}Rf'(0) &= \lim_{h \rightarrow 0} \frac{(0+h)(\sqrt{0+h} - \sqrt{0+h+1}) - 0}{h} \\ &= \lim_{h \rightarrow 0} \sqrt{h} - \sqrt{0+h+1} \\ &= -\sqrt{1} = -1\end{aligned}$$

$\therefore f$ is differentiable at $x = 0$

5. If $y = 3x^5 + 4x^4 + 2x + 3$, then

- (A) $y_4 = 0$ (B) $y_5 = 0$ (C) $y_6 = 0$ (D) none of these

Ans. (C) $y_6 = 0$

$\Rightarrow$ Here, the maximum power of x is 5.

$$\therefore y_6 = 0$$

6. If $y = \sqrt{(1-x)(1+x)}$, then

- (A) $(1-x^2)\frac{dy}{dx} - xy = 0$ (B) $(1-x^2)\frac{dy}{dx} + xy = 0$
(C) $(1-x^2)\frac{dy}{dx} - 2xy = 0$ (D) $(1-x^2)\frac{dy}{dx} + 2xy = 0$

Ans. (B) $(1-x^2)\frac{dy}{dx} + xy = 0$

$$\Rightarrow y^2 = (1-x^2) \Rightarrow 2y\frac{dy}{dx} = -2x \text{ or } \frac{dy}{dx} = \frac{-x}{y} = \frac{-xy}{(1-x^2)} \text{ or } (1-x^2)\frac{dy}{dx} + xy = 0$$

7. If $y = \frac{\tan x + \cot x}{\tan x - \cot x}$, then $\frac{dy}{dx} = \dots\dots\dots$

- (A) $2\tan 2x \sec 2x$ (B) $\tan 2x \sec 2x$ (C) $-\tan 2x \sec 2x$ (D) $-2\tan 2x \sec 2x$

Ans. (D) $-2\tan 2x \sec 2x$

$$\Rightarrow y = \frac{\tan x + \cot x}{\tan x - \cot x} = -\frac{1 + \tan^2 x}{1 - \tan^2 x} = -\sec 2x$$

$$\Rightarrow \frac{dy}{dx} = -2 \tan 2x \sec 2x$$

8. $\frac{d}{dx} \left[\log \sqrt{\sin \sqrt{e^x}} \right] = \dots\dots$

(A) $\frac{1}{4} e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$

(B) $e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$

(C) $\frac{1}{4} e^x \cot (e^x)$

(D) $\frac{1}{2} e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$

Ans. (A) $\frac{1}{4} e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$

$$\Rightarrow \frac{d}{dx} [\log \sqrt{\sin \sqrt{e^x}}] = \frac{d}{dx} \left[\frac{1}{2} \log (\sin \sqrt{e^x}) \right]$$

$$= \frac{1}{2} \cot \sqrt{e^x} \cdot \frac{1}{2\sqrt{e^x}} e^x = \frac{1}{4} e^{\frac{x}{2}} \cot \left(e^{\frac{x}{2}} \right)$$

9. $\left[\frac{d}{dx} \sec^{-1} x \right]_{x=-3} = \dots\dots\dots$

(A) $\frac{1}{\sqrt{x^2 - 1}}$

(B) $\frac{-1}{\sqrt{x^2 - 1}}$

(C) $\frac{1}{6\sqrt{2}}$

(D) $-\frac{1}{6\sqrt{2}}$

Ans. (C) $\frac{1}{6\sqrt{2}}$

$$\Rightarrow \frac{d}{dx} \sec^{-1} x = \frac{1}{|x| \sqrt{x^2 - 1}}$$

$$\therefore \left[\frac{d}{dx} \sec^{-1} x \right]_{x=-3} = \frac{1}{|-3| \sqrt{(-3)^2 - 1}}$$

$$= \frac{1}{3\sqrt{9-1}}$$

$$= \frac{1}{3\sqrt{8}}$$

$$= \frac{1}{6\sqrt{2}}$$

10. $\frac{d}{dx} \sin^{-1} (\cos 2x) = \dots\dots \left(0 < x < \frac{\pi}{2} \right)$

(A) 2

(B) -2

(C) 1

(D) 5

Ans. (B) -2

$$\Rightarrow \frac{d}{dx} \sin^{-1} (\cos 2x)$$

$$= \frac{d}{dx} \sin^{-1} \left[\sin \left(\frac{\pi}{2} - 2x \right) \right] \quad \left(0 < x < \frac{\pi}{2} \right)$$

$$= \frac{d}{dx} \left(\frac{\pi}{2} - 2x \right) = 0 - 2(1) = -2$$

11. $x^{\cos y} + y^{\cos x} = 5$ then

(A) $x = 0, y = 0 \Rightarrow y' = 0$

(B) $x = 0, y = 1 \Rightarrow y' = 0$

(C) $x = y = 1 \Rightarrow y' = -1$

(D) $x = 1, y = 0 \Rightarrow y' = 1$

Ans. (C) $x = y = 1 \Rightarrow y' = -1$

$$\Rightarrow x^{\cos y} + y^{\cos x} = 5$$

$$\therefore e^{\cos y \cdot \log_e x} + e^{\cos x \cdot \log_e y} = 5$$

Differentiate with respect to x on both sides,

$$\therefore e^{\cos y \cdot \log_e x} \left[\frac{\cos y}{x} - \log_e x \sin y \cdot \frac{dy}{dx} \right] + e^{\cos x \cdot \log_e y} \left[\frac{\cos x}{y} \cdot \frac{dy}{dx} - \sin x \cdot \log_e y \right] = 0$$

Now taking $x = y = 1$

$$\therefore (\cos 1 - 0) + \left(\cos 1 \frac{dy}{dx} - 0 \right) = 0$$

$$\therefore \frac{dy}{dx} = -1 \Rightarrow y' = -1$$

12. $\frac{d^n}{dx^n}(\log x) = \dots\dots\dots$

- (A) $\frac{(n-1)!}{x^n}$ (B) $\frac{n!}{x^n}$ (C) $\frac{(n-2)!}{x^n}$ (D) $(-1)^{n-1} \frac{(n-1)!}{x^n}$

Ans. (D) $(-1)^{n-1} \frac{(n-1)!}{x^n}$

$$\Rightarrow \text{Here } y = \log x$$

Differentiate w.r. to x ,

$$\therefore y_1 = \frac{1}{x}, y_2 = \frac{-1}{x^2}, y_3 = \frac{2}{x^3}, \dots\dots\dots$$

$$y_n = (-1)^{n-1} \frac{(n-1)!}{x^n}$$

13. $\frac{d}{dx}(\tan x^\circ) = \dots\dots\dots$

- (A) $\pi \sec^2 x^\circ$ (B) $\frac{180}{\pi} \sec^2 x^\circ$ (C) $\sec^2 x^\circ$ (D) $\frac{\pi}{180} \sec^2 x^\circ$

Ans. (D) $\frac{\pi}{180} \sec^2 x^\circ$

$$\Rightarrow \frac{d}{dx}(\tan x^\circ) = \frac{d}{dx} \tan\left(\frac{\pi x}{180}\right)$$

$$= \sec^2\left(\frac{\pi x}{180}\right) \cdot \frac{\pi}{180} \frac{dx}{dx}$$

$$= \frac{\pi}{180} \sec^2(x^\circ)$$

14. If $e^y(x+1) = 1$, then what is the value of $\frac{d^2 y}{dx^2} = \dots\dots\dots$

- (A) $-\left(\frac{dy}{dx}\right)^2$ (B) $\left(\frac{dy}{dx}\right)$ (C) $-\left(\frac{dy}{dx}\right)$ (D) $\left(\frac{dy}{dx}\right)^2$

Ans. (D) $\left(\frac{dy}{dx}\right)^2$

$$\Rightarrow e^y(x+1) = 1$$

Differentiating with respect to x , $\frac{d}{dx} [e^y(x+1)] = 1$

$$(x+1) \frac{d}{dx}(e^y) + e^y \frac{d}{dx}(x+1) = \frac{d}{dx}(1)$$

$$\therefore (x+1) e^y \cdot \frac{dy}{dx} + e^y = 0$$

$$\therefore \frac{dy}{dx} = -\frac{1}{(x+1)} \quad \dots\dots(i)$$

Again differentiating with respect to x ,

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(-\frac{1}{x+1} \right)$$

$$\therefore \frac{d^2y}{dx^2} = \frac{(x+1)(0) - (-1)}{(x+1)^2} = \frac{1}{(x+1)^2}$$

$$\therefore \frac{d^2y}{dx^2} = \left(\frac{dy}{dx} \right)^2 \quad (\because \text{From (i)})$$

15. $f(x) = \begin{cases} kx^2 & \text{if } x \leq 2 \\ 3 & \text{if } x > 2 \end{cases}$ If f is continuous at $x = 2$, then the value of k is

(A) $\frac{4}{3}$

(B) $\frac{3}{4}$

(C) 4

(D) 3

Ans. (B) $\frac{3}{4}$

→ f is continuous at $x = 2$

$$\therefore \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$\therefore \lim_{x \rightarrow 2^+} 3 = \lim_{x \rightarrow 2^-} kx^2$$

$$\therefore 3 = \lim_{h \rightarrow 0} k(2-h)^2 \quad (\because \text{Taking } x = 2-h)$$

$$\therefore 3 = \lim_{h \rightarrow 0} k(4-4h+h^2)$$

$$\therefore 3 = 4k$$

$$\therefore k = \frac{3}{4}$$

16. $\frac{d}{dx} e^{x^3} = \dots\dots$

(A) e^{x^3}

(B) $x^3 \cdot e^{x^3-1}$

(C) $3x^2 e^{x^3}$

(D) $e^{x^3} \log_3 e$

Ans. (C) $3x^2 e^{x^3}$

$$\rightarrow \frac{d}{dx} (e^{x^3}) = e^{x^3} \cdot \frac{d}{dx} (x^3)$$

$$= e^{x^3} \cdot 3x^2$$

$$= 3x^2 \cdot e^{x^3}$$

17. If $f(x) = \log_a n x^n$ then $f'(1) = \dots\dots$

(A) $\frac{1}{x \log a}$

(B) 0

(C) $\frac{1}{\log a}$

(D) $\frac{n}{\log n}$

Ans. (C) $\frac{1}{\log a}$

$\Rightarrow$ Here $f(x) = \log_a nx^n$
 $= \frac{\log_e x^n}{\log_e a^n}$
 $= \frac{n \log x}{n \log a} = \frac{\log x}{\log a}$

$\therefore f'(x) = \frac{d}{dx} \left(\frac{\log x}{\log a} \right)$
 $= \frac{1}{\log a} \frac{d}{dx} (\log x)$
 $= \frac{1}{\log a} \cdot \frac{1}{x}$

$\therefore f'(1) = \frac{1}{\log a} \times 1 = \frac{1}{\log a}$

18. $\frac{d}{dx} (x^{2x}) = \dots\dots$

(A) $x^{2x}(1 + 2\log x)$

(B) $x^{2x}(2 + \log x)$

(C) $2x^{2x}(1 + \log x)$

(D) does not exist

Ans. (C) $2x^{2x}(1 + \log x)$

$\Rightarrow$ Let, $y = x^{2x}$

$\therefore$ taking log on both the sides

$\log y = \log x^{2x} = 2x \log x$

$\therefore$ Differentiating with respect to x on both the sides

$\frac{1}{y} \frac{dy}{dx} = 2x \cdot \frac{1}{x} + 2 \cdot \log x$

$\therefore \frac{dy}{dx} = y(2 + 2\log x) = 2 \cdot x^{2x} \cdot (1 + \log x)$

19. $\frac{d}{dx} (\operatorname{cosec}^{-1} e^x) = \dots\dots\dots$

(A) $\frac{-1}{\sqrt{e^{2x} - 1}}$

(B) $\sin^{-1}(e^x)$

(C) $\frac{1}{\sqrt{e^{2x} - 1}}$

(D) $\frac{-e^x}{\sqrt{e^{2x} - 1}}$

Ans. (A) $\frac{-1}{\sqrt{e^{2x} - 1}}$

$\Rightarrow \frac{d}{dx} (\operatorname{cosec}^{-1} e^x) = \frac{-1}{|e^x| \sqrt{(e^x)^2 - 1}} \frac{d}{dx} (e^x)$
 $= \frac{-1}{e^x \sqrt{e^{2x} - 1}} \cdot e^x \quad (\because e^x > 0 \Rightarrow |e^x| = e^x)$
 $= \frac{-1}{\sqrt{e^{2x} - 1}}$

20. If $y = 100e^{2x} + 200e^{-2x}$ and $\frac{d^2y}{dx^2} = ay$, then $a = \dots\dots$

(A) 0

(B) -4

(C) 2

(D) 4

Ans. (D) 4

$\Rightarrow y = 100e^{2x} + 200e^{-2x}$

$$\therefore \frac{dy}{dx} = 100 \cdot e^{2x} \cdot 2 + 200 \cdot e^{-2x} \cdot (-2)$$

$$\begin{aligned}\therefore \frac{d^2y}{dx^2} &= 100 \cdot 2 \cdot e^{2x} \cdot 2 + 200(-2) \cdot e^{-2x} \cdot (-2) \\ &= 4(100 \cdot e^{2x} + 200 \cdot e^{-2x})\end{aligned}$$

$$\therefore \frac{d^2y}{dx^2} = 4y = ay$$

$$\therefore a = 4$$

21. If $x + 1 = e^{-y}$, then $\frac{d^2y}{dx^2} = \dots\dots\dots$

- (A) $\frac{dy}{dx}$ (B) $\left(\frac{dy}{dx}\right)^2$ (C) $\left(\frac{dy}{dx}\right)^3$ (D) $-\frac{dy}{dx}$

Ans. (B) $\left(\frac{dy}{dx}\right)^2$

Here $x + 1 = e^{-y}$

$$\therefore x + 1 = \frac{1}{e^y}$$

$$\therefore e^y = \frac{1}{x + 1}$$

$\therefore$ Differentiating w.r. to x ,

$$e^y \frac{dy}{dx} = -\frac{1}{(x+1)^2} \frac{d}{dx}(x+1) \left(\because \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2} \right)$$

$$\therefore \frac{1}{x+1} \frac{dy}{dx} = -\frac{1}{(x+1)^2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{x+1} \quad \dots (1)$$

Again differentiating w.r. to x ,

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}\left(-\frac{1}{x+1}\right) \\ &= -\frac{d}{dx}\left(\frac{1}{x+1}\right) \\ &= -1 \times \frac{-1}{(x+1)^2} \frac{d}{dx}(x+1) \\ &= \frac{1}{(x+1)^2} (1) \\ &= \left(\frac{-1}{x+1}\right)^2 \\ &= \left(\frac{dy}{dx}\right)^2 \quad (\because (1))\end{aligned}$$

22. If $x = \sqrt{10^{\sin^{-1}t}}$, $y = \sqrt{10^{\cos^{-1}t}}$, then $\frac{dy}{dx} = \dots\dots\dots$

- (A) $-\frac{y}{x}$ (B) $\frac{y}{x}$ (C) 0 (D) $-\frac{x}{y}$

Ans. (A) $-\frac{y}{x}$

➡ $x = \sqrt{10^{\sin^{-1}t}}, y = \sqrt{10^{\cos^{-1}t}}$

$\therefore x \cdot y = \sqrt{10^{\sin^{-1}t}} \cdot \sqrt{10^{\cos^{-1}t}}$

$\therefore xy = \sqrt{10^{\sin^{-1}t + \cos^{-1}t}}$

$\therefore xy = \sqrt{10^{\frac{\pi}{2}}} \quad \left(\because \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2} \right)$

$\therefore xy = \text{constant}$

$\therefore$ Differentiating with respect to x

$\therefore x \cdot \frac{dy}{dx} + y = 0$

$\therefore \frac{dy}{dx} = -\frac{y}{x}$

23. $\left\{ \frac{d}{dx}(x^x + x^{x+1} + x^{x+2}) \right\}_{x=e} = \dots\dots\dots$

- (A) $e^e(2e^2 + 4e + 3)$ (B) $e^e(1 + e^2 + 2e)$ (C) $e^e(3e^2 + 2e + 2)$ (D) $e^e(1 + 4e + 2e^2)$

Ans. (A) $e^e(2e^2 + 4e + 3)$

➡ $\frac{d}{dx}(x^x + x^{x+1} + x^{x+2})$

$= \frac{d}{dx}(x^x + x^{x \cdot \infty} \cdot x + x^x \cdot x^2)$

$= \frac{d}{dx}x^x(1 + x + x^2)$

$= x^x \cdot \frac{d}{dx}(1 + x + x^2) + (1 + x + x^2)^{\infty} \cdot \frac{d}{dx}(x^{x \cdot \infty})$

$= x^{x \cdot \infty}(1 + 2x) + (1 + x + x^2)^{\infty} \cdot x^x \cdot (1 + \log x)$

$= x^{x \cdot \infty}\{1 + 2x + (1 + x + x^2)^{\infty} \cdot (1 + \log x)\}$

$\therefore$ Taking $x = e \left\{ \frac{d}{dx}(x^x + x^{x+1} + x^{x+2}) \right\}_{x=e}$

$= e^e\{1 + 2e + (1 + e + e^2)^{\infty} \cdot (1 + \log e)\}$

$= e^e\{1 + 2e + (1 + e + e^2)^{\infty} \cdot 2\}$

$= e^e\{1 + 2e + 2 + 2e + 2e^2\}$

$= e^e\{2e^2 + 4e + 3\}$

Section B

- Write the answer of the following questions. [Each carries 2 Marks]

[18]

1. Find the values of k so that the function f is continuous at the indicated point in :

$f(x) = \begin{cases} kx + 1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases} \text{ at } x = \pi$

➡ The function f is continuous at $x = \pi$.

$\therefore \lim_{x \rightarrow \pi^+} f(x) = \lim_{x \rightarrow \pi^-} f(x)$

$$\therefore \lim_{x \rightarrow \pi^+} \cos x = \lim_{x \rightarrow \pi^-} kx + 1$$

$$\therefore \lim_{h \rightarrow 0} \cos(\pi + h) = \lim_{h \rightarrow 0} k(\pi - h) + 1$$

$$\therefore \cos \pi = k\pi + 1$$

$$\therefore -1 = k\pi + 1$$

$$\therefore k = -\frac{2}{\pi}$$

2. Examine that $\sin|x|$ is a continuous function.

➡ Let $f(x) = |x|$ and $g(x) = \sin x$

$$\therefore (g \circ f)(x) = g[f(x)]$$

$$= g|x|$$

$$= \sin|x|$$

$\therefore f(x)$ is a modulus function and $g(x)$ is a trigonometric function. so, they are continuous.

$\therefore$ Their composite function $(g \circ f)(x)$ is also continuous.

$\therefore \sin|x|$ is a continuous function.

3. Find $\frac{dy}{dx}$ in the following : $ax + by^2 = \cos y$

➡ $ax + by^2 = \cos y$

Differentiating both sides w.r. to x , we get,

$$\therefore a + 2by \frac{dy}{dx} = -\sin y \frac{dy}{dx}$$

$$\therefore 2by \frac{dy}{dx} + \sin y \frac{dy}{dx} = -a$$

$$\therefore \frac{dy}{dx} [2by + \sin y] = -a$$

$$\therefore \frac{dy}{dx} = \frac{-a}{2by + \sin y}$$

4. Find $\frac{dy}{dx}$ in the following : $\sin^2 x + \cos^2 y = 1$

➡ $\sin^2 x + \cos^2 y = 1$

Differentiating both sides w.r. to x , we get,

$$\therefore 2\sin x \left(\frac{d}{dx} \sin x \right) + 2\cos y \left(\frac{d}{dx} \cos y \right) = 0$$

$$\therefore 2\sin x \cdot \cos x - 2\cos y \sin y \frac{dy}{dx} = 0$$

$$\therefore \sin 2x - \sin 2y \frac{dy}{dx} = 0$$

$$\therefore \sin 2y \frac{dy}{dx} = \sin 2x$$

$$\therefore \frac{dy}{dx} = \frac{\sin 2x}{\sin 2y}$$

5. Differentiate the following w.r.t. x : $\frac{e^x}{\sin x}$

➡ $y = \frac{e^x}{\sin x}$

On differentiating both sides w.r. to x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{e^x}{\sin x} \right) \\ &= \frac{\sin x \cdot \frac{d}{dx}(e^x) - e^x \cdot \frac{d}{dx}(\sin x)}{\sin^2 x} \quad (\text{By quotient rule}) \\ &= \frac{\sin x \cdot e^x - e^x \cdot \cos x}{\sin^2 x} \\ &= \frac{e^x (\sin x - \cos x)}{\sin^2 x}, \text{ where } x \neq n\pi, n \in \mathbb{I}.\end{aligned}$$

6. Differentiate the following w.r.t. x : e^{x^3}

➡ $y = e^{x^3}$

On differentiating both sides w.r. to x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (e^{x^3}) \\ &= e^{x^3} \cdot \frac{d}{dx} (x^3) \\ &= e^{x^3} \cdot 3x^2 \\ &= 3x^2 \cdot e^{x^3}\end{aligned}$$

7. Differentiate the following w.r.t. x : $e^{\sin^{-1} x}$

➡ $y = e^{\sin^{-1} x}$

On differentiating both sides w.r. to x , we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (e^{\sin^{-1} x}) \\ &= e^{\sin^{-1} x} \cdot \frac{d}{dx} (\sin^{-1} x) \quad (\text{Using chain rule}) \\ &= e^{\sin^{-1} x} \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}} \quad x \in (-1, 1)\end{aligned}$$

8. Differentiate the following w.r.t. x : $e^x + e^{x^2} + \dots + e^{x^5}$

➡ $y = e^x + e^{x^2} + \dots + e^{x^5}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} [e^x + e^{x^2} + e^{x^3} + e^{x^4} + e^{x^5}] \\ &= \frac{d}{dx} (e^x) + \frac{d}{dx} (e^{x^2}) + \frac{d}{dx} (e^{x^3}) + \frac{d}{dx} (e^{x^4}) + \frac{d}{dx} (e^{x^5})\end{aligned}$$

$$= e^x + e^{x^2} \frac{d}{dx}(x^2) + e^{x^3} \frac{d}{dx}(x^3) + e^{x^4} \frac{d}{dx}(x^4) + e^{x^5} \frac{d}{dx}(x^5)$$

$$\therefore \frac{dy}{dx} = e^x + 2x \cdot e^{x^2} + 3x^2 \cdot e^{x^3} + 4x^3 \cdot e^{x^4} + 5x^4 \cdot e^{x^5}$$

9. Differentiate the function with respect to x : $\cos(\sqrt{x})$

$$\Rightarrow y = \cos(\sqrt{x})$$

$$\frac{dy}{dx} = \frac{d}{dx} (\cos(\sqrt{x}))$$

$$\frac{dy}{dx} = -\sin(\sqrt{x}) \cdot \frac{d}{dx} (\sqrt{x})$$

$$= -\sin(\sqrt{x}) \cdot \left(\frac{1}{2} x^{\frac{1}{2}-1} \right)$$

$$= \frac{-\sin(\sqrt{x})}{2\sqrt{x}}$$

Section C

● Write the answer of the following questions. [Each carries 3 Marks]

[9]

10. For what value of λ is the function defined by $f(x) = \begin{cases} \lambda(x^2 - 2x), & \text{if } x \leq 0 \\ 4x + 1, & \text{if } x > 0 \end{cases}$ continuous at $x = 0$? What about continuity at $x = 1$?

$$\Rightarrow f(x) = \begin{cases} \lambda(x^2 - 2x), & x \leq 0 \\ 4x + 1, & x > 0 \end{cases} \text{ is continuous at } x = 0.$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \lambda(x^2 - 2x)$$

$$= \lim_{h \rightarrow 0} \lambda[(0 - h)^2 - 2(0 - h)]$$

$$= \lim_{h \rightarrow 0} \lambda[h^2 + 2h] = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 4x + 1$$

$$= \lim_{h \rightarrow 0} 4(0 + h) + 1$$

$$= \lim_{h \rightarrow 0} 4h + 1$$

$$= 1$$

The function f is continuous at $x = 0$.

$$\therefore \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$\therefore 0 = 1$ which is not possible.

$\therefore$ For no value of λ , f is continuous at $x = 0$.

At $x = 1$,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 4x + 1 = 5$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 4x + 1 = 5$$

$\therefore f$ is continuous at $x = 1$ for any value of λ .

11. Find the values of a and b such that the function defined by $f(x) = \begin{cases} 5 & , \text{ If } x \leq 2 \\ ax + b, & \text{ If } 2 < x < 10 \\ 21 & , \text{ If } x \geq 10 \end{cases}$ is a continuous function.

➡ For $x < 2$, $f(x) = 5$ is a constant function, so it is continuous.

For $2 < x < 10$, $f(x) = ax + b$ is a polynomial, so it is continuous.

For $x > 10$, $f(x) = 21$ is constant function, so it is continuous.

Now we check the continuity of $f(x)$ at $x = 2$ and $x = 10$.

The function $f(x)$ is continuous at $x = 2$.

$$\therefore \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\therefore \lim_{x \rightarrow 2^-} 5 = \lim_{x \rightarrow 2^+} ax + b$$

$$\therefore 5 = \lim_{h \rightarrow 0} a(2 + h) + b$$

$$\therefore 5 = 2a + b \quad \dots\dots\dots(i)$$

A function $f(x)$ is continuous at $x = 10$.

$$\therefore \lim_{x \rightarrow 10^-} f(x) = \lim_{x \rightarrow 10^+} f(x)$$

$$\therefore \lim_{x \rightarrow 10^-} ax + b = \lim_{x \rightarrow 10^+} 21$$

$$\therefore \lim_{h \rightarrow 0} a(10 + h) + b = 21$$

$$\therefore 10a + b = 21 \quad \dots\dots\dots(ii)$$

Here (ii) - (i) $\Rightarrow 8a = 16 \Rightarrow a = 2$

and (i) $\Rightarrow 5 = 4 + b \Rightarrow b = 1$

12. Find all the points of discontinuity of f defined by $f(x) = |x| - |x + 1|$.

➡ $f(x) = |x| - |x + 1|$

when $x < -1$, $f(x) = -x + (x + 1) = 1$

when $-1 \leq x < 0$, $f(x) = -x - (x + 1) = -2x - 1$

when $x \geq 0$, $f(x) = x - (x + 1) = -1$

$$\text{Thus, } f(x) = \begin{cases} 1, & x < -1 \\ -2x - 1, & -1 \leq x < 0 \\ -1, & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} 1 = 1$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} -2x - 1 = -2(-1) - 1 = 1$$

and $f(-1) = -2(-1) - 1 = 1$

$$\text{Thus, } \lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x) = f(-1)$$

$\therefore$ The function $f(x)$ is continuous at $x = -1$.

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$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -2x - 1 = -1$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} -1 = -1 \text{ and } f(0) = -1$$

Thus, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$

$\therefore$ The function $f(x)$ is continuous at $x = 0$.

For $x < -1$, $f(x) = 1$ is a constant function so it is continuous.

For $-1 < x < 0$, $f(x) = -2x - 1$ is a polynomial function so it is continuous.

For $x > 0$, $f(x) = -1$ is constant function so it is continuous.

Also, f is continuous at $x = -1$ and $x = 0$.

$\therefore f(x)$ is continuous for $\forall x \in \mathbb{R}$.

$\therefore$ There is no points of discontinuity.